

FOCAL-K: Interaction-Focused Bimanual Diffusion with Object-Centric State and Keyframe Transition Retrieval

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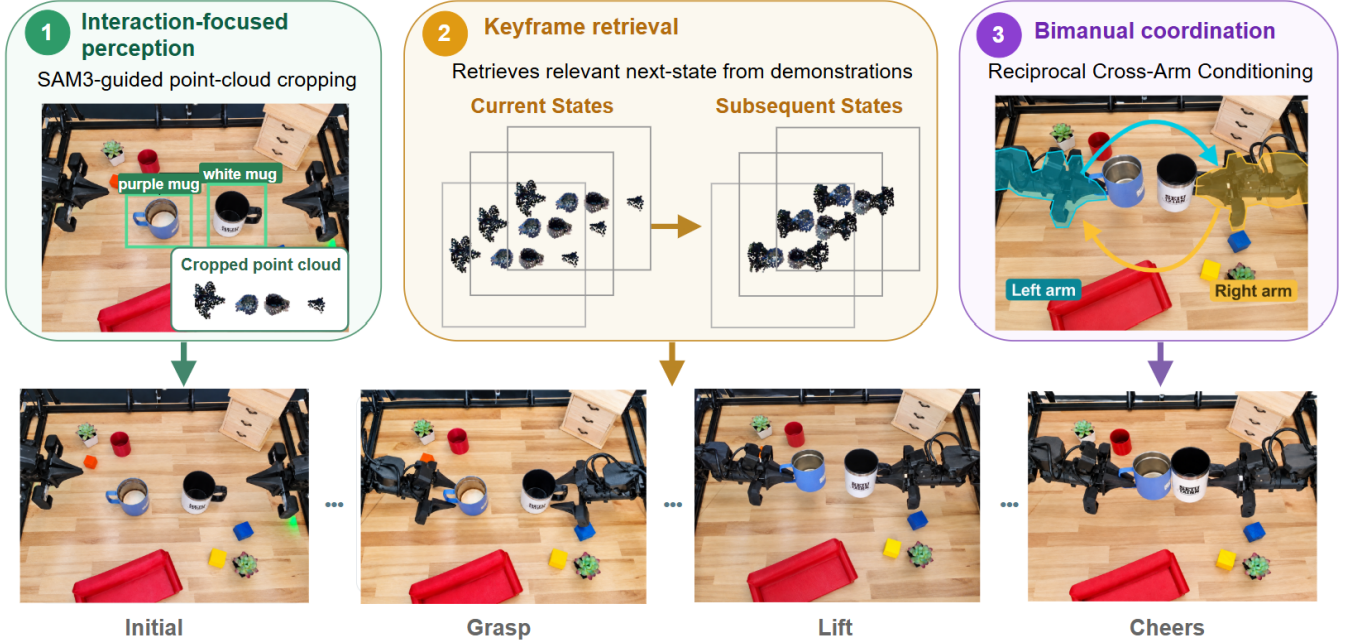


Fig. 1: Real-world *Mug Cheers* task featuring FOCAL-K, requiring precise thin-handle grasps learned from expert demonstrations, coordinated dual-mug lifting, and in-air contact. (1) Interaction-focused point-cloud cropping. (2) Keyframe retrieval with successor state. (3) Reciprocal cross-arm conditioning during denoising. The lower sequence shows representative keyframes across task stages during policy execution on ALOHA [1]; dots indicate omitted frames.

Abstract—Bimanual manipulation requires reasoning over task-relevant geometry and task progression while coordinating two arms under evolving scene conditions. 3D diffusion policies provide a strong framework for imitation-based manipulation, yet point clouds can leave task-relevant structure implicit, keyframe-based task progression often relies on manual annotation, and dual-arm control introduces synchronization and role-assignment challenges. We introduce FOCAL-K, an interaction-focused 3D diffusion policy that addresses these challenges through object-grounded interaction geometry and explicit object state, automatically selected keyframes retrieved by robot-object state to provide transition context beyond demonstrated configurations, and reciprocal cross-arm conditioning throughout diffusion denoising. Together, these components provide structured spatial and transition context while reinforcing inter-arm coordination within continuous action generation. Across ten RoboTwin 2.0 tasks, FOCAL-K achieves 69.0%/24.5% average success under Clean/Randomized conditions, compared with 50.3%/3.1% for Bimanual DP3 and 59.0%/19.1% for $\pi_{0.5}$; across three real-world ALOHA tasks, it achieves 60.0%/68.3%, outperforming both baselines under both settings. Additional results and videos are available at <https://focal-k.github.io/>.

I. INTRODUCTION

Bimanual manipulation requires coordinated action as object configurations, contacts, and arm roles evolve throughout

a task. Each arm’s motion depends on the current task state and the other arm, particularly during contact, transfer, and simultaneous manipulation [1]–[5].

Diffusion policies provide a strong framework for multimodal continuous control through iterative denoising [6], [7]. Recent work increasingly structures the information available to this process. *For perception*, 3D policies encode scene geometry with point clouds [8], while object-centric, task-focused, multi-view, and language-grounded approaches emphasize task-relevant geometry [9], [10]. Maintaining this focus as objects move and visibility changes remains an important representation problem. *For task progression*, memory-based methods retrieve prior demonstrations, while keyframe-based approaches often use annotated salient states to structure behavior [11]–[14], highlighting the value of demonstrated transitions as structured context beyond the immediate observation. *For bimanual coordination*, prior work structures cross-arm dependencies through roles, relational representations, or feature exchange [15]–[17], providing alternatives to relying solely on implicit coordination through joint action prediction.

Building on these directions, we present **FOCAL-K**, a bimanual 3D diffusion policy targeting these three aspects

of action generation. Its **Structured Observation Representation** tracks task objects across multiple views and organizes object–gripper geometry with world-frame object state, maintaining interaction focus as the scene evolves. The **Keyframe Database (KFDB)** automatically selects keyframes from demonstrations and retrieves immediate-successor records from similar robot–object configurations, providing demonstrated transition context. **Bimanual Coordination Diffusion** introduces explicit reciprocal cross-arm conditioning at each denoising step, where each arm’s intermediate action estimate modulates the other arm’s prediction. Figure 2 shows the overall architecture.

We evaluate FOCAL-K across ten RoboTwin 2.0 tasks [18] and three real-world tasks on an ALOHA platform. FOCAL-K achieves higher task-average success than Bimanual DP3 and $\pi_{0.5}$ under both Clean and Randomized conditions in simulation and the real world. In summary, our main contributions are:

- **Structured Observation Representation.** A multi-view, interaction-focused 3D representation that structures object–gripper geometry and object state for bimanual manipulation.
- **Keyframe Database Retrieval.** An automatically constructed nonparametric database that provides demonstrated transition context for action generation.
- **Bimanual Coordination Diffusion.** Reciprocal cross-arm conditioning that uses each arm’s intermediate action estimate to modulate the other throughout diffusion denoising.
- **Empirical Evaluation.** Across ten RoboTwin 2.0 and three real-world ALOHA tasks, FOCAL-K achieves higher task-average success than Bimanual DP3 and $\pi_{0.5}$ under both Clean and Randomized conditions.

II. RELATED WORK

A. 3D Representations for Robot Manipulation

Effective manipulation requires capturing scene geometry and task-relevant structure. PointNet and PointNet++ learn geometric features from unordered point clouds [19], [20], underpinning 3D visuomotor policies conditioned on workspace geometry [8]. Object-centric and language-grounded methods focus observations on task-relevant objects and regions, with collision-aware approaches further distinguishing obstacles [9], [10], [21], [22]. Vision–language–action models such as $\pi_{0.5}$ instead learn visuolinguistic representations from RGB and language at scale [23]. FOCAL-K further structures 3D observations through multi-view, task-focused tracking with occlusion handling, providing explicit object state for policy learning and keyframe retrieval.

B. Retrieval, Keyframes, and Memory

Prior experience can contextualize both the current state and subsequent task progression. Demonstration retrieval uses neighboring states and actions for prediction, conditioning, or data augmentation [11], [13], [14], [24];

keyframe methods use salient states as annotated change-points, intermediate targets, or motion anchors [12], [25], [26]; and memory-based approaches retain subtask-annotated keyframes to preserve long-horizon task history [27]. FOCAL-K instead structures demonstrations around automatically identified keyframe transitions, using robot proprioception and explicit object state to retrieve relevant progression context for continuous action generation at configurations beyond those directly demonstrated.

C. Bimanual Coordination

Bimanual manipulation expands manipulation capabilities but requires coordinating interdependent actions as contacts and arm roles evolve. Prior work structures coordination through asymmetric constraints, acting–stabilizing roles, robot–object relations, action chunking, or cross-arm feature exchange [1], [15]–[17], [28], [29]. Diffusion-based bimanual policies further incorporate large-scale pretraining, relational structure, future-state prediction, and task-specific objectives [30]–[33], while recent methods supervise keyposes, object motion, or relational predictions to guide action generation [34], [35]. FOCAL-K augments implicit joint coordination with reciprocal cross-arm conditioning, using each arm’s intermediate action estimate to refine the other’s prediction throughout denoising, thereby learning coordinated, role-flexible behavior directly from demonstrations.

III. PROBLEM STATEMENT

We study single-task bimanual imitation learning from N expert demonstrations, $\mathcal{D} = \{\tau_n\}_{n=1}^N$, containing synchronized observations and actions. At time t , the robot receives synchronized multi-view RGB-D observations $\mathcal{R}_t$ and bimanual proprioception $\mathbf{q}_t = [\mathbf{q}_t^L; \mathbf{q}_t^R] \in \mathbb{R}^{14}$. The action $\mathbf{a}_t = [\mathbf{a}_t^L; \mathbf{a}_t^R] \in \mathbb{R}^{14}$ specifies the next absolute joint and gripper target for both arms. Each task has a fixed instruction g that identifies its task-relevant objects. We train a separate policy for each task; the instruction is used for object selection within that task. We denote the observation at time step t as $\mathbf{o}_t = (\mathcal{R}_t, \mathbf{q}_t)$. The policy conditions on an observation history $\mathbf{O}_t$ comprising the latest T_o observations and predicts a distribution over an H -step coordinated action chunk:

$$p_\theta(\mathbf{A}_t | \mathbf{O}_t), \quad \mathbf{O}_t = \mathbf{o}_{t-T_o+1:t}, \quad \mathbf{A}_t = \mathbf{a}_{t:t+H-1}. \quad (1)$$

The objective is to reproduce coordinated expert behavior under variations in object configuration and observed scene conditions. We assume fixed camera calibration, robot embodiment, action space, and task semantics, with no additional expert supervision beyond the training demonstrations.

IV. METHOD

A. Structured Observation Representation

Problem Statement. Point-cloud-based diffusion policies such as DP3 represent the observation at time t as

$$\mathcal{O}_t^{\text{DP3}} = \{S_t^r, P_t^s\}, \quad (2)$$

where S_t^r is the robot state and P_t^s a scene-level point-cloud embedding. The policy must therefore infer task-relevant

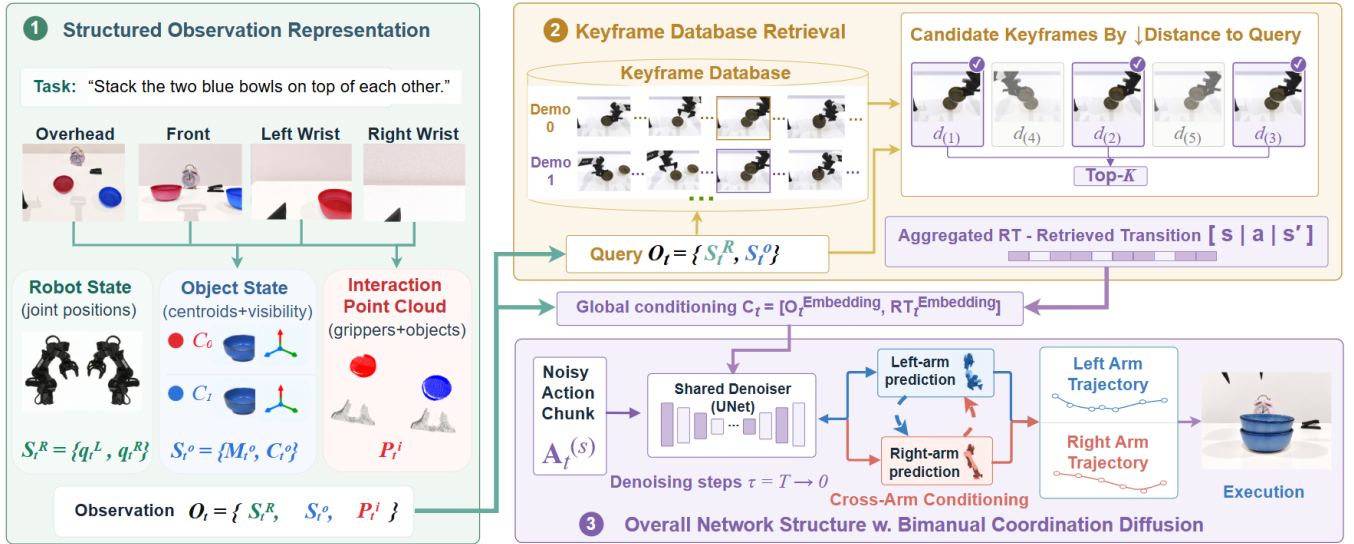


Fig. 2: **FOCAL-K overview.** (1) **Structured Observation Representation** extracts multi-view object–grripper geometry and structured object state, which together with robot proprioception form the policy observation. (2) **Keyframe Database Retrieval** matches the current robot–object state to keyframes selected from training demonstrations and aggregates the top- K successor transitions as retrieval context. (3) **Bimanual Coordination Diffusion** conditions on observation and retrieval context, with reciprocal cross-arm conditioning during denoising to generate coordinated trajectories.

objects, their spatial states, and object–grripper interactions from a single unordered representation amid scene clutter. Language-guided object-centric methods address this by isolating task-relevant objects, but object state and interaction geometry remain coupled within the point cloud. Furthermore, single-view RGB-D observations are vulnerable to occlusion, particularly in bimanual settings where two arms can simultaneously obscure objects and contact regions.

We therefore formulate manipulation perception around three questions:

- 1) **Which objects are task-relevant?**
- 2) **Where are the goal objects located?**
- 3) **What is the object–grripper interaction geometry?**

We address this decomposition with a structured observation observation constructed from interaction-aware multi-view RGB-D inputs:

$$\mathcal{O}_t = \{S_t^r, S_t^o, P_t^i\}, \quad S_t^r = \{\mathbf{q}_t^L, \mathbf{q}_t^R\}, \quad (3)$$

where $\mathbf{q}_t^L$ and $\mathbf{q}_t^R$ are the left- and right-arm joint states, $S_t^o = (\mathbf{M}_t, \mathbf{C}_t)$ contains canonically ordered object visibility masks and world-frame centroids, and P_t^i is an interaction-centric point-cloud embedding of goal-object geometry, gripper geometry, and their local interactions.

This factorization explicitly represents global object state in S_t^o while specializing P_t^i for local interaction geometry.

Language-Guided Semantic Object Selection. To identify which objects are task-relevant, we use off-the-shelf foundation models independently of policy learning. GPT-4o mini extracts goal-object categories from the task instruction (e.g., ‘stack the two blue bowls’ → ‘blue bowl’). We then collage the overhead, front, and two wrist views and use Gemini-3-Flash-Preview to ground all goal objects in a single inference pass, selected from preliminary comparisons with SAM3, Grounding DINO, and Florence-2.

The resulting per-view masks are associated via a common bird’s-eye-view (BEV) projection to match object instances across views, establish persistent identities, and initialize SAM3 tracking, yielding per-object, per-view masks, bounding boxes, and 2D centroids throughout the episode.

Occlusion-Aware Multi-View Registration and Centroid Estimation.

Multi-View Centroid Estimation. We consider two strategies for estimating world-frame object centroids. The first uses *multi-view semantic point-cloud fusion*, which back-projects and fuses per-view segmented object points. While preserving detailed object geometry, its centroid is sensitive to segmentation drift and visibility changes under occlusion: changes in the reconstructed surface can shift the estimate despite little object motion. For policy conditioning, we observe that a consistently object-anchored centroid is preferable to one that approximates the instantaneous geometric center but exhibits spurious frame-to-frame shifts.

Our final estimator therefore prioritizes temporal consistency. Tracked bounding boxes from the approximately orthogonal overhead and front views provide complementary geometric constraints for isolating the object region from the scene point cloud, applied sequentially when both are available. The centroid $\mathbf{c}_t^i$ is obtained from the resulting constrained 3D region. This provides a stable object-associated spatial reference without requiring precise surface reconstruction at each observation.

Occlusion Handling and Crop Recovery. For each object and view, visibility is assessed from the current mask area relative to its maximum observed area. The 2D per-view centroid is continuously updated from the current mask, while the bounding-box extent is updated only under sufficient visibility. For the overhead and front views, offsets from the centroid to the box boundaries are retained; under partial occlusion, these offsets are applied around the current

centroid, allowing the crop to follow the tracked object without shrinking with the partially occluded mask.

Structured Object-State Construction. The estimated world-frame centroids form $S_t^o = (\mathbf{M}_t, \mathbf{C}_t)$, where $\mathbf{C}_t$ contains canonically ordered object centroids and $\mathbf{M}_t$ their visibility states. The policy is explicitly conditioned on this global object state $\mathcal{O}_t$ (see Eq. 3), separating explicit object location from the local interaction geometry encoded by P_t^i .

Across our experiments, reliable centroid estimates are associated with higher policy performance and lower run-to-run variability, whereas unsuccessful rollouts under severe occlusion or limited-view observations often exhibit centroid drift. These results support explicit object-centroid state as an effective policy signal while underscoring the importance of accurate, temporally consistent localization.

This structured observation state is further leveraged by our Keyframe Database component, where (S_t^r, S_t^o) jointly defines the retrieval key for demonstrated keyframes with similar robot-object configurations.

B. Keyframe Database Retrieval

To provide examples of how manipulation proceeds, we match the current robot-object state to similar keyframes in training demonstrations and retrieve their successor-keyframe information as context for action generation.

Keyframe database. We automatically identify keyframe candidates at gripper changes or bimanual stops, capturing interaction changes and pauses without manual annotation:

$$\mathcal{K}_{\text{cand}} = \{t : \|\mathbf{v}_t\|_\infty \leq \epsilon_v \vee \mathbf{b}_t \neq \mathbf{b}_{t-1}\}, \quad (4)$$

where $\mathbf{v}_t$ contains both arms' per-frame joint-position differences as velocity estimates, ϵ_v is the stopping threshold, and $\mathbf{b}_t$ contains the two binary gripper states. Initial and terminal frames are retained.

For each nonterminal keyframe i , let i^+ denote its immediate successor within the same demonstration. We store the transition record

$$\mathbf{h}_i = [\mathbf{q}_i; \mathbf{a}_i; \mathbf{q}_{i^+}], \quad (5)$$

containing the robot state, its associated action, and the subsequent keyframe state. We build

$$\mathcal{D}_{\text{KF}} = \{(\mathbf{k}_i, \mathbf{h}_i)\}_{i=1}^{N_{\text{KF}}}, \quad (6)$$

$$\mathbf{k}_i = [\text{Norm}_q(\mathbf{q}_i); \mathbf{M}_i; \text{vec}(\text{Norm}_C(\mathbf{C}_i))],$$

where N_{KF} counts nonterminal keyframes and vec flattens ordered centroids. Norm_q and Norm_C normalize robot states and centroids using training statistics shared across keys and queries. Transition records remain unnormalized.

Successor-keyframe retrieval. At time t , the latest robot and object states (S_t^r, S_t^o) form the query $\mathbf{k}_t$ using the key representation in Eq. 6. The query uses the latest tracked state without concatenating the policy's observation history or accessing future rollout observations. Each matched key retrieves a transition record containing the demonstrated robot state, its associated action, and the subsequent keyframe state. These records provide examples of how demonstrations proceed from similar configurations.

Let $\mathcal{N}_t$ index the K nearest eligible database entries. We aggregate their transition records, assigning greater weight to closer configurations:

$$d_i = \|\mathbf{k}_t - \mathbf{k}_i\|_2^2, \quad \bar{d}_i = \frac{d_i - d_{\min}}{\max(d_{\max} - d_{\min}, \delta)}, \quad (7)$$

$$w_i = \frac{\exp(-\lambda \bar{d}_i)}{\sum_{j \in \mathcal{N}_t} \exp(-\lambda \bar{d}_j)}, \quad \mathbf{z}_t^{\text{ret}} = \sum_{i \in \mathcal{N}_t} w_i \mathbf{h}_i,$$

where $d_{\min}$ and $d_{\max}$ are the minimum and maximum distances within $\mathcal{N}_t$, $\lambda > 0$ controls weight sharpness, and $\delta > 0$ prevents division by zero. The resulting $\mathbf{z}_t^{\text{ret}}$ is the aggregated retrieved transition used for policy conditioning. During training, we exclude all entries from the query demonstration to prevent trajectory-level leakage. At evaluation, the database contains only records from training demonstrations.

Policy conditioning. A learned keyframe encoder f_{KF} maps the aggregated transition record $\mathbf{z}_t^{\text{ret}}$ to a retrieval embedding $\mathbf{e}_t^{\text{KF}}$. We concatenate this embedding with the observation-history feature $\mathbf{c}_t^{\text{obs}}$ to form the denoiser condition $\mathbf{c}_t$:

$$\mathbf{e}_t^{\text{KF}} = f_{\text{KF}}(\mathbf{z}_t^{\text{ret}}), \quad \mathbf{c}_t = [\mathbf{c}_t^{\text{obs}}; \mathbf{e}_t^{\text{KF}}]. \quad (8)$$

The retrieved transitions provide contextual guidance for action generation. Retrieved actions are not executed directly, and retrieved states are not used as waypoint targets. We train the keyframe encoder and diffusion policy jointly using the diffusion objective described next.

C. Bimanual Coordination Diffusion

Bimanual manipulation requires each arm's motion to account for the other arm's behavior. Although a shared denoiser can learn this dependence implicitly, we introduce an explicit reciprocal adjustment of its action predictions. Our hypothesis is that conditioning each arm's adjustment on the other arm's proposed motion provides a useful inductive bias for coordination.

Intermediate action estimation. We adopt the action prediction formulation of DP3 [8]. The prediction target is the normalized action chunk before diffusion noise is added, $\mathbf{A}_t^{(0)} \in \mathbb{R}^{H \times 14}$. Given its noisy counterpart $\mathbf{A}_t^{(s)}$ at diffusion step s and the combined condition $\mathbf{c}_t$, the shared temporal U-Net F_θ produces an intermediate action estimate:

$$\hat{\mathbf{A}}_t^{(0|s)} = F_\theta(\mathbf{A}_t^{(s)}, s, \mathbf{c}_t) = [\hat{\mathbf{A}}_t^{L, (0|s)}; \hat{\mathbf{A}}_t^{R, (0|s)}]. \quad (9)$$

The superscript $(0|s)$ denotes an estimate of $\mathbf{A}_t^{(0)}$ from diffusion step s . Brackets concatenate the two $H \times 7$ arm predictions along the action dimension.

Reciprocal cross-arm modulation. We use each arm's preliminary action prediction to condition a FiLM-style affine adjustment of the opposite arm's prediction [36]. For $(u, v) \in \{(L, R), (R, L)\}$, a direction-specific network $f_{v \rightarrow u}$ generates modulation parameters:

$$(\gamma_{t,s}^u, \beta_{t,s}^u) = f_{v \rightarrow u}(\hat{\mathbf{A}}_t^{v, (0|s)}), \quad (10)$$

$$\tilde{\mathbf{A}}_t^{u, (0|s)} = \hat{\mathbf{A}}_t^{u, (0|s)} \odot (1 + \tanh \gamma_{t,s}^u) + \beta_{t,s}^u.$$

TABLE I: **Simulation results.** Success rate (%) averaged over three training seeds under benchmark-defined Clean and Randomized test conditions, with all policies trained on Clean demonstrations. Each run is evaluated on the same 50 test episodes per condition. We additionally study training on Randomized demonstrations for six of the ten tasks; Table II shows substantially improved Randomized-condition performance for FOCAL-K relative to Bimanual DP3. Bold denotes the numerically best result, including ties.

Task	$\pi_{0.5}$		Bimanual DP3		FOCAL-K	
	Clean	Randomized	Clean	Randomized	Clean	Randomized
Beat Block Hammer	82.7	20.7	76.7	8.0	88.0	36.0
Place Can Basket	74.7	12.0	76.0	3.3	88.7	22.0
Pick Dual Bottles	86.7	15.3	64.0	4.0	88.7	30.7
Handover Block	64.0	17.3	76.0	0.0	92.0	22.0
Stack Bowls Two	90.7	76.0	85.3	13.3	92.0	60.7
Stack Blocks Two	45.3	19.3	24.0	0.0	45.3	20.0
Scan Object	36.7	5.3	30.0	1.3	63.3	6.0
Hanging Mug	34.0	4.0	17.3	0.7	47.3	13.3
Place Bread Skillet	40.7	12.0	36.0	0.0	48.7	17.3
Place Dual Shoes	34.7	9.3	17.3	0.0	36.0	17.3
Task average	59.0	19.1	50.3	3.1	69.0	24.5

Here, $\gamma_{t,s}^u, \beta_{t,s}^u \in \mathbb{R}^{H \times 7}$ parameterize scaling and bias, $\odot$ denotes element-wise multiplication, and $\tilde{\mathbf{A}}$ denotes the adjusted prediction. Both directions use the unmodulated predictions. Each network operates separately at each horizon position, with parameters shared across positions and diffusion steps.

V. EXPERIMENTS

We evaluate FOCAL-K against Bimanual DP3 [8] as the 3D diffusion-policy baseline across ten RoboTwin 2.0 tasks under benchmark-defined *Clean* and *Randomized* settings. We also compare with the pretrained VLA $\pi_{0.5}$ [23] as a strong reference despite its distinct RGB-based representation and large-scale pretraining. Ablations examine perception, retrieval, and cross-arm conditioning, followed by evaluation on an ALOHA platform.

Denoising and training. At each reverse-diffusion step, the reciprocally conditioned left- and right-arm predictions are concatenated into the joint bimanual action estimate $\tilde{\mathbf{A}}_t^{(0|s)} = [\tilde{\mathbf{A}}_t^{L,(0|s)}; \tilde{\mathbf{A}}_t^{R,(0|s)}]$, which is passed to the standard DP3 reverse-diffusion update. We retain DP3’s forward process and jointly train the denoiser and cross-arm conditioning networks with

$$\mathcal{L}_{\text{diff}} = \mathbb{E} \left[\frac{1}{14H} \left\| \tilde{\mathbf{A}}_t^{(0|s)} - \mathbf{A}_t^{(0)} \right\|_F^2 \right], \quad (11)$$

where H is the prediction horizon and $14H = 2 \times 7H$ is the joint trajectory dimension for two 7-D arms. The expectation is over training samples, diffusion steps, and forward-process noise. No auxiliary coordination loss is used.

A. Simulation Setup

Benchmark and tasks.

We evaluate on ten bimanual tasks from RoboTwin 2.0 [18] using the ALOHA-AgileX embodiment: *Beat Block Hammer*, *Place Can Basket*, *Pick Dual Bottles*, *Handover Block*, *Stack Bowls Two*, *Stack Blocks Two*, *Scan Object*, *Hanging Mug*, *Place Bread Skillet*, and *Place Dual Shoes*.

TABLE II: **Effect of randomized training.** Success rate (%) across six tasks when all three methods are trained and evaluated under Randomized conditions. Bold denotes the numerically best result, including ties.

Task	DP3	$\pi_{0.5}$	FOCAL-K
Beat Block Hammer	2	66	72
Place Can Basket	6	40	40
Pick Dual Bottles	4	40	48
Handover Block	2	26	84
Stack Bowls Two	6	88	64
Stack Blocks Two	4	34	18
Average	4.0	49.0	54.3

These tasks cover tool interaction, target placement, simultaneous dual-object manipulation, inter-arm handover, and geometric stacking.

Observation inputs. FOCAL-K and Bimanual DP3 use point clouds derived from four RGB-D views: front, overhead, left wrist, and right wrist. $\pi_{0.5}$ uses RGB images from the corresponding four views.

Training and evaluation. FOCAL-K and Bimanual DP3 are trained on the same 50 Clean demonstrations per task for 3,000 epochs, with checkpoints validated every 500 epochs. The pretrained $\pi_{0.5}$ is fine-tuned on the same demonstrations for 100k steps, with checkpoints validated every 10k steps. For each method and task, we select the checkpoint with the highest held-out Clean validation performance before evaluation. Three fixed training seeds are used, with disjoint Clean and Randomized test sets shared across methods and seeds. Each run evaluates 50 episodes per setting using the benchmark’s binary success criteria, and success rates are averaged across the three runs. Ablations follow the same protocol unless stated otherwise.

B. Simulation Results

Table I reports results across ten tasks. Under Clean, FOCAL-K achieves a task-average success rate of 69.0%, exceeding $\pi_{0.5}$ and Bimanual DP3 by 10.0 and 18.7 percentage points, respectively. It achieves the highest mean success rate, including ties, in across all ten tasks.

TABLE III: **Component ablation.** Mean success rate (%) over three runs under the Clean (C) and Randomized (R) settings. *Struct. Obs.* and *Co. Diff.* denote structured observation and coordination diffusion, respectively. Best results are shown in bold, including ties.

Variant	Pick Dual Bottles		Stack Blocks Two	
	C	R	C	R
DP3	64.0	4.0	24.0	0.0
+ Struct. Obs.	62.7	16.0	20.0	9.3
+ KFDB	88.0	0.0	28.7	0.0
+ Co. Diff.	84.0	2.0	28.0	0.0
+ KFDB + Co. Diff.	88.7	2.7	29.3	11.3
+ Struct. Obs. + KFDB	88.0	29.3	26.7	11.3
+ Struct. Obs. + Co. Diff.	85.3	27.3	27.3	11.3
All (FOCAL-K)	88.7	30.7	45.3	20.0

TABLE IV: **Ablation on object-centroid conditioning.** Success rate (%) using one seed when object centroids are excluded, used only for KFDB retrieval, or additionally included in the policy observation.

Task	No Centroid		Centroid in KFDB		KFDB + Observation	
	C	R	C	R	C	R
Handover Block	70	52	96	60	100	60
Pick Dual Bottles	60	44	90	42	88	48
Stack Bowls Two	83	44	88	60	88	72

Under Randomized, FOCAL-K achieves 24.5% mean success, compared with 19.1% for $\pi_{0.5}$ and 3.1% for Bimanual DP3. It outperforms Bimanual DP3 on every task and achieves the highest reported rate on nine tasks. It also obtains nonzero success on all four tasks where Bimanual DP3 reports zero success. Since training uses clean demonstrations and checkpoint selection uses clean validation, these results suggest improved robustness to evaluation-time randomization relative to the DP3 backbone.

The advantage over $\pi_{0.5}$ is more task-dependent under Randomized, with the average margin narrowing from 10.0 to 5.4 percentage points. On *Stack Bowls Two*, for example, $\pi_{0.5}$ achieves 76.0%, compared with 60.7% for FOCAL-K. The decrease from 69.0% under Clean to 24.5% under Randomized should be interpreted in the context of training exclusively on Clean demonstrations, such that Randomized evaluation introduces substantial distribution shift. When trained on Randomized demonstrations, FOCAL-K recovers substantially higher success on several tasks, indicating that demonstration coverage is an important factor in robustness to scene variation.

C. Ablation Studies

Component contributions. We evaluate all eight combinations of structured observation (Struct. Obs.), KFDB retrieval, and coordination diffusion (Co. Diff.) on *Pick Dual Bottles* and *Stack Blocks Two*, and These tasks test simultaneous object manipulation and precise stacking, respectively. All variants use the Bimanual DP3 backbone. With structured observation disabled and KFDB retrieval active, the policy receives the original full-scene point cloud. The retrieval query replaces the ordered object-centroid state with the mean XYZ coordinate of the workspace point cloud.

On *Pick Dual Bottles*, adding KFDB retrieval or coordination diffusion raises clean success more than 20% (Table III).

TABLE V: **Keyframe database ablations.** Clean success (%) using one seed per configuration. Top: query variants at $K = 4$. Bottom: varying K with robot state and object centroids. Bold: best in each section.

Query / K	Pick Dual Bottles	Stack Bowls Two
Robot state only	82	72
+ Point-cloud centroid	84	80
+ Object centroids	88	82
$K = 2$	80	88
$K = 3$	84	90
$K = 4$	88	82
$K = 5$	78	80

TABLE VI: **Ablation on coordination diffusion.** Mean Clean success (%) over three seeds on Handover Block. Left: modulation placement under the standard objective. Right: objectives with post-U-Net modulation.

Modulation placement		Training objective	
Variant	Success	Variant	Success
No modulation	62	Standard	84
Pre-U-Net	68	+ Agreement	60
Mid-U-Net	48	+ Direction	14
Post-U-Net	84	+ Magnitude	40

These variants reach less than 2% under randomized. Adding structured observation raises their randomized rates more than 25%. On *Stack Blocks Two*, the full model reaches 45.3% under clean and 20.0% under randomized, exceeding every partial configuration in both settings.

Object-centroid conditioning. We ablate where explicit object-centroid state is introduced by progressively adding it to KFDB retrieval and the policy observation. We compare no centroid conditioning, centroid-conditioned KFDB retrieval, and centroid conditioning in both KFDB and the observation vector. All other settings remain fixed. Results are reported in Table IV.

Keyframe database design. We ablate the keyframe database query representation and neighborhood size. Neighbors are ranked by squared ℓ_2 distance. During training, all records originating from the query demonstration are excluded before nearest-neighbor search to prevent same-trajectory retrieval. All other settings remain fixed.

The upper section of Table V compares three queries at $K = 4$: robot state alone, with a single interaction-cloud centroid (without SAM3 tracking), and with ordered object centroids and availability indicators. The complete query improves success over robot state alone by 6 and 10 percentage points on *Pick Dual Bottles* and *Stack Bowls Two*, respectively. The lower section varies K with the complete query fixed. Success peaks at $K = 4$ and $K = 3$ for the two tasks, respectively, and declines at $K = 5$ for both. We select K per task on the validation set.

Coordination-diffusion design. With structured observation and KFDB retrieval disabled, we compare the unmodulated backbone with modulation placed before, within, or after the U-Net. Each result averages three training seeds. We evaluate checkpoints after a fixed number of training epochs, without validation-based checkpoint selection. Post-U-Net modulation has the highest reported mean success

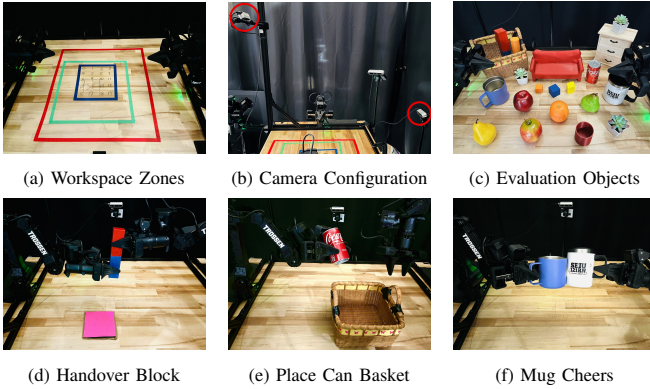


Fig. 3: **Real-world experimental setup and tasks.** Top: (a) nested workspace zones, (b) overhead and front RGB-D camera configuration, and (c) complete evaluation object set. Bottom: (d) *Handover Block*, (e) *Place Can Basket*, and (f) *Mug Cheers*. The blue, green, and red boundaries in (a) denote progressively larger workspace regions used for task-object initialization and environmental randomization.

rate among these placements (Table VI). At that placement, the standard diffusion objective also has a higher mean rate than variants with prediction-agreement, direction-alignment, or magnitude-synchronization losses. We use post-U-Net modulation and the standard objective in the full model.

D. Real-World Experiments

Platform and Tasks. We evaluate Bimanual DP3, our method, and $\pi_{0.5}$ on a Trossen ALOHA platform with dual WidowX AI arms across three real-world tasks. *Handover Block* tests sequential inter-arm coordination through object transfer followed by target-conditioned placement. *Place Can Basket* tests asymmetric multi-stage manipulation, with one arm placing a can into the basket before the other grasps and lifts the basket. *Mug Cheers* tests precise dual-arm coordination, with both arms establishing stable handle grasps and aligning the mugs for controlled in-air contact. Together, they provide representative probes of inter-arm transfer, asymmetric multi-stage manipulation, and precise dual-arm spatial coordination.

Workspace and Execution Configuration. For perception, we use one overhead and one front-facing Intel RealSense D435 RGB-D camera. The workspace is defined by robot reachability and the reliable depth operating ranges of both cameras, ensuring stable RGB-D observations while permitting spatial variation. The block, can, and mugs are initialized within the inner blue zone; the basket and target pads within the green zone; and distractors may extend throughout the larger red zone. See Fig. 3 for the real-world setup and tasks.

Bimanual DP3 and our method use depth-reconstructed point clouds processed with RealSense SDK parameter tuning and a multi-stage filtering pipeline for point-cloud cleaning and artifact removal, while $\pi_{0.5}$ operates directly on RGB observations. All cameras and the ALOHA control loop operate at 15 Hz, while perception runs asynchronously in parallel with action execution, with approximately 250–300 ms latency from camera input to the processed, cropped point cloud. We empirically tune each policy’s rollout horizon to balance execution continuity with observation freshness:

TABLE VII: **Real-world task success.** Success rate (%) under Clean (C) and Randomized (R) conditions.

Task	Bimanual DP3		$\pi_{0.5}$		FOCAL-K	
	C	R	C	R	C	R
Handover Block	60	20	20	20	60	70
Place Can Basket	60	50	80	75	70	85
Mug Cheers	50	10	0	0	50	50
Average	56.7	33.3	33.3	31.6	60.0	68.3

Bimanual DP3 and our method use 6-step action chunks, while $\pi_{0.5}$ uses a 10-step horizon.

Training and Evaluation Protocol. We collect 50 successful expert demonstrations per task using the ALOHA leader arms and train all methods on the same dataset. Bimanual DP3 and our method are trained for 3000 epochs, with checkpoints evaluated every 500 epochs; $\pi_{0.5}$ is trained for 100k steps, with checkpoints evaluated every 10k steps. Based on the simulation sweep, we carry two candidate checkpoints per policy into real-world validation: 500 and 1000 epochs for Bimanual DP3 and ours, and 20k and 100k steps for $\pi_{0.5}$. The better-performing checkpoint on real-world validation trials is used for final evaluation; across our tasks, this selects 500 epochs for Bimanual DP3 and ours and 20k steps for $\pi_{0.5}$.

For real-world evaluation, we design experiments to assess two dimensions of performance: *spatial generalization* and *environmental robustness*. Clean trials include both seen and unseen object placements, while Randomized trials introduce semantic distractors, clutter, object-height and lighting variations, and combined perturbations. Each evaluation round comprises 10 Clean and 10 Randomized trials, with success rates averaged across repeated rounds using task-specific binary completion criteria.

Results. Table VII summarizes task success rates.

Spatial generalization. All methods perform reliably at seen locations. $\pi_{0.5}$ generalizes strongly to unseen positions, while our method retains substantial success through explicit object grounding. Larger out-of-distribution spatial shifts remain challenging for the DP3-based policies.

Environmental robustness. Bimanual DP3 degrades sharply with increasing clutter, consistent with dilution of task-relevant geometry within its fixed 1024-point representation. Our interaction-centric representation largely preserves performance, achieving robustness comparable to $\pi_{0.5}$ under clutter, although both degrade under dense clutter or severe occlusion. Language-conditioned grounding further enables our method to resolve appearance-defined targets that geometry-only DP3 cannot distinguish and provides more reliable target selection than $\pi_{0.5}$ under the evaluated semantic distractors. All methods remain robust to lighting variation, while our method exhibits the least degradation under height variation.

Task-specific motion adaptation. On *Mug Cheers*, our method more reliably acquires the demonstrated handle grasp, approach, and in-air contact behavior than $\pi_{0.5}$, demonstrating effective adaptation to specialized motion primitives from task-specific demonstrations.

Overall, Bimanual DP3 effectively acquires task-specific behaviors but is more sensitive to scene variation, while $\pi_{0.5}$ provides strong pretrained spatial generalization but is less reliable on the evaluated inter-arm coordination and specialized motion behaviors. Our method combines task-specific adaptation with improved robustness to clutter and semantic ambiguity using only 50 task demonstrations.

VI. LIMITATIONS

FOCAL-K depends on reliable object-state estimation: grounding errors, occlusion, partial observations, and view-point changes can degrade object geometry and cause centroid drift. Across our simulation and real-world evaluations, centroid errors beyond approximately 2.5 cm are associated with reduced task success, although sensitivity and the effective threshold vary by task. Object 3D completion and full-occlusion-robust tracking are promising directions for preventing centroid drift under severe or complete occlusion. KFDB also depends on keyframe quality and demonstration coverage; poorly represented configurations may provide less informative transition context. Finally, we do not explicitly model or evaluate collision avoidance.

VII. CONCLUSION

We present FOCAL-K, an interaction-focused 3D diffusion policy that structures bimanual action generation through multi-view object–gripper state, automatic keyframe selection and retrieval of demonstrated transitions, and reciprocal cross-arm conditioning. Across ten simulated and three real-world tasks, FOCAL-K consistently improves task-average success over bimanual DP3 and $\pi_{0.5}$ under both Clean and Randomized conditions. We hope these studies motivate future work on more effective structures for 3D observation, adaptive keyframe selection and retrieval for long-horizon manipulation, and tighter coordination of interdependent arm actions.

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